Adam Rida

AI Researcher & Machine Learning Engineer
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Portfolio: adrida.github.io
Email: adamrida.ra@gmail.com
Github – Google Scholar

Experience

DeepRecall — Paris, France
Solo Founder — *Dec 2025 – present*

- **$100k ARR in 3 months (solo founder, bootstrapped)**; accepted into **Station F F/AI** (recommendation-only; 20 startups selected globally; backed by OpenAI, Anthropic, Google, Meta, Microsoft, Sequoia); enterprise pilot with a **$15B global e-commerce marketplace**
- Developed **FUSE**, a cross-modal representation fusion algorithm combining text and multi-image embeddings for robust multimodal similarity search
- Built the **largest structured product-safety dataset globally (200k+ regulatory alerts)** by reverse-engineering and scraping national safety alert systems across multiple jurisdictions
- Designed a **multimodal retrieval architecture** leveraging embedding models and vector search, exposed as an **MCP-compatible tool server** enabling agents to perform automated product safety investigations
- Designed a **learned deferral framework** routing predictions between classical ML surrogates and LLMs based on sample difficulty, with no measurable degradation in prediction quality at **80%+ lower inference cost**

Autoplay AI — Paris / New York City
Head of AI - Founding Team — *Jan 2025 – Dec 2025*

- Developed real-time ML models analyzing millions of behavioral events from session replays to generate per-user cognitive and intent scores
- Built a hybrid ML + LLM system detecting user hesitation, intent and knowledge gaps from behavioral signals
- Identified struggling-user cohorts using unsupervised clustering, enabling targeted UX improvements
- Designed **TERRA** (Task-Event Representation & Reasoning Architecture), a knowledge graph-based behavioral ontology modeling user intent and actions across software interactions, used to ground reasoning agents and reduce costs

Entrepreneur First - SynthAI — Paris, France
Co-founder and CTO — *May 2024 – Dec 2024*

- Built an AI copilot for investment thesis generation deployed to **100+ private equity funds**
- Developed GraphRAG-based multi-document analysis pipelines for financial document synthesis
- Engineered a custom PDF parsing system outperforming off-the-shelf extractors; company acquired by a customer

MFG Labs — Paris, France
Data Scientist II — *Nov 2023 – Apr 2024*

- Applied machine learning to supply chain demand forecasting for **Hermès**
- Designed optimization strategies reducing stockouts while maintaining demand coverage

AXA / Sorbonne University – CNRS — Paris, France
AI Research Scientist – Ph.D. Candidate — *Oct 2022 – Oct 2023*

- Researched Explainable AI and differential explanations within AXA's R&D team
- Authored a paper accepted at the **ECML/PKDD 2023 DynXAI Workshop**

Société Générale CIB — Paris, France
ML Data Scientist — *Sept 2021 – Sept 2022*

- Built ML-based time series models predicting transaction volumes and counts in a scarce-data environment (>20M rows/month)
- Developed scalable data pipelines using PySpark and Kedro

Rebellion Research — NYC, USA
Machine Learning Quantitative Researcher Intern — *Feb 2021 – Sept 2021*

- Developed deep reinforcement learning trading strategies achieving **21% annualized return / 1.8 Sharpe**
- Engineered features from SEC 13-F filings improving model signal quality

Qantev Paris, France
Data Scientist Intern & Software Engineer *Apr 2021 – Oct 2021*

- Built NLP models for medical insurance claim classification and optimized data processing pipelines

Education

Sorbonne University – CNRS - LIP6 – AXA Paris, France
Ph.D. Candidate in AI *2022 – 2023*

- Specialized in Explainable AI (XAI) focusing on differential explanations between models.
- Part of the AXA-Sorbonne University Trustworthy and Responsible AI Lab (TRAIL), supervised by Christophe Marsala, Marie-Jeanne Lesot and Xavier Renard.
- Transitioned to industry after one year after a personal emergency.

CY Cergy Paris Université - CY Tech (ex EISTI - École d'Ingénieur) Cergy, France
Master's degree in Applied Mathematics and Data Science – (Diplôme d'Ingénieur) *2017 – 2022*

- Ranked in the top 8% with a grade of 16/20 (Grade A).

Ecole 42 Paris, France
Peer-to-peer Programming Program *2017 – 2020*

- Ranked 24th out of 900 applicants in a competitive one-month programming session

Lycée Charlemagne Paris 4, France
Baccalauréat (Honors) *2016*

- Among top Parisians high schools - French Baccalauréat with Honors, Mention Bien

Skills

- **AI & ML**: Agent Architectures, Multimodal Retrieval, Embeddings, XAI, RAG pipelines, GraphRAG, DRL, Transformers, NLP
- **Programming**: Python (most proficient language), R, Java, C, Scala, SQL
- **Tools & Frameworks**: Scikit-learn, pandas, numpy, PyTorch, TensorFlow, Kedro, PySpark, LangChain, Pinecone, Gradio
- **Languages**: French: Native — English: Fluent (TOEIC 965) — Spanish: Conversational

Publications & Projects *(all links, more details and access to code at adrida.github.io)*

- **Dynamic Interpretability for Model Comparison via Decision Rules**, Adam Rida, Marie-Jeanne Lesot, Xavier Renard, Christophe Marsala; ECML/PKDD 2023 DynXAI Explainable Artificial Intelligence: From Static to Dynamic Workshop
- **Deep Reinforcement Learning & Feature Extraction For Constructing Alpha Generating Equity Portfolios**, Alexander Fleiss*, Amrith Kumaar*, Adam Rida*, Junsup Shin, Xinying Lai, Vivian Fang, Jialiang Chen, Ang Li:: SSRN (800+ downloads *:equal contributors)
- **Temporal Augmented Retrieval (TAR) - Dynamic RAG** (Open Source Project - 27+ stars on Github)
- Essays on Rebellion Research: XAI, Crypto, Human–AI Coordination.
- Web3: Projects with 3K+ users and multi-million market caps (details on request).